Functoriality of cocones under a map of spans commutes with postcomposition

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:conemap-funct' (source lines 941-951, Lemma 7.4.10); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. Two context spans are retained: the author's definition of a map of spans, which the statement quantifies over, and the author's construction of its action on cocones together with the numbered display defining the induced homotopy.

Editorial note. The retained counter values (chapter 7, section 4, theorem offset 7) make the retained definition print as Definition 7.4.8 and the counted lemma as Lemma 7.4.10, matching the original book numbering; theorem-class environments share one counter through aliascnt.

Editorial note. The single external reference points at the naturality display of section 7.3 that the retained context cites; it is a navigation pointer only.

Editorial scope. One original result of the HoTT Book, counted once: functoriality of cocones under a map of spans commutes with postcomposition (Lemma 7.4.10).

Allowed standard prerequisites (editorial).

- Cocones under a span. A cocone under a span (A <-f- C -g-> B) with vertex D is a triple (i,j,h) with i : A -> D, j : B -> D and h : prd z:C i(f z) = j(g z).. The statement is about the two ways of transporting such a triple along t : D -> E.
- Maps of spans and their action on cocones. A map of spans (alpha,beta,gamma,phi,psi) acts on cocones by composing the legs and filling the square with the composite displayed in the author's numbered equation.. Both composites in the statement go through that action, and the proof compares the two induced homotopies.
- ap preserves path concatenation. ap t of a composite of paths is the composite of the aps.. It is the last step of the proof at source line 974.

Source reference. Original equation 7.3.4: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1329–1329. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the

counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

We now observe that the maps from each type to its n-truncation assemble into a map of spans, in the following sense.

Definition 7.4.8. Let

$$\mathscr{D} = \quad \begin{array}{ccc} C & \xrightarrow{g} & B \\ {\scriptstyle f}\downarrow & & \\ A & & \end{array} \qquad \text{and} \qquad \mathscr{D}' = \quad \begin{array}{ccc} C' & \xrightarrow{g'} & B' \\ {\scriptstyle f'}\downarrow & & \\ A' & & \end{array}$$

be spans. A **map of spans** $\mathscr{D} \to \mathscr{D}'$ consists of functions $\alpha : A \to A'$, $\beta : B \to B'$, and $\gamma : C \to C'$ and homotopies $\phi : \alpha \circ f \sim f' \circ \gamma$ and $\psi : \beta \circ g \sim g' \circ \gamma$.

Thus, for any span $\mathscr{D}$, we have a map of spans $|-|_n^{\mathscr{D}} : \mathscr{D} \to \|\mathscr{D}\|_n$ consisting of $|-|_n^A$, $|-|_n^B$, $|-|_n^C$, and the naturality homotopies nat_n^f and nat_n^g from (7.3.4).

We also need to know that maps of spans behave functorially. Namely, if $(\alpha, \beta, \gamma, \phi, \psi) : \mathscr{D} \to \mathscr{D}'$ is a map of spans and D any type, then we have

$$\begin{cases} \mathsf{cocone}_{\mathscr{D}'}(D) & \longrightarrow & \mathsf{cocone}_{\mathscr{D}}(D) \\ (i, j, h) & \longmapsto & (i \circ \alpha, j \circ \beta, k) \end{cases}$$

where $k : \prod_{(z:C)} i(\alpha(f(z))) = j(\beta(g(z)))$ is the composite

$$i(\alpha(f(z))) \xlongequal{\mathsf{ap}_i(\phi)} i(f'(\gamma(z))) \xlongequal{h(\gamma(z))} j(g'(\gamma(z))) \xlongequal{\mathsf{ap}_j(\psi)} j(\beta(g(z))). \tag{7.4.9}$$

We denote this cocone by $(i, j, h) \circ (\alpha, \beta, \gamma, \phi, \psi)$.

Lemma 7.4.10. *Given $(\alpha, \beta, \gamma, \phi, \psi) : \mathscr{D} \to \mathscr{D}'$ and $t : D \to E$, the following diagram commutes:*

$$\begin{array}{ccc} \mathsf{cocone}_{\mathscr{D}'}(D) & \xrightarrow{t \circ -} & \mathsf{cocone}_{\mathscr{D}'}(E) \\ \downarrow & & \downarrow \\ \mathsf{cocone}_{\mathscr{D}}(D) & \xrightarrow[t \circ -]{} & \mathsf{cocone}_{\mathscr{D}}(E) \end{array}$$

Proof. Given $(i, j, h) : \mathsf{cocone}_{\mathscr{D}'}(D)$, note that both composites yield a cocone whose first two components are $t \circ i \circ \alpha$ and $t \circ j \circ \beta$. Thus, it remains to verify that the homotopies agree. For the top-right composite, the homotopy is (7.4.9) with (i, j, h) replaced by $(t \circ i, t \circ j, \mathsf{ap}_t \circ h)$:

$$t\, i\, \alpha\, f\, z \xlongequal{\mathsf{ap}_{t \circ i}(\phi)} t\, i\, f'\, \gamma\, z \xlongequal{\mathsf{ap}_t(h(\gamma(z)))} t\, j\, g'\, \gamma\, z \xlongequal{\mathsf{ap}_{t \circ j}(\psi)} t\, j\, \beta\, g\, z$$

(For brevity, we are omitting the parentheses around the arguments of functions.) On the other hand, for the left-bottom composite, the homotopy is ap_t applied to (7.4.9). Since ap respects path-concatenation, this is equal to

$$t\, i\, \alpha\, f\, z \xlongequal{\mathsf{ap}_t(\mathsf{ap}_i(\phi))} t\, i\, f'\, \gamma\, z \xlongequal{\mathsf{ap}_t(h(\gamma(z)))} t\, j\, g'\, \gamma\, z \xlongequal{\mathsf{ap}_t(\mathsf{ap}_j(\psi))} t\, j\, \beta\, g\, z.$$

But $\mathsf{ap}_t \circ \mathsf{ap}_i = \mathsf{ap}_{t \circ i}$ and similarly for j, so these two homotopies are equal. □